

Companion XXIII

The Formation of the First Black Holes in the Relaxation-Driven Cyclic Cosmology Seed Channels, Early Growth, and High-Redshift Quasars

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Abstract

The discovery of supermassive black holes (SMBHs) with masses exceeding $10^8 M_\odot$ at redshifts $z > 10$ poses a major challenge to the standard Λ CDM cosmology. The rapid assembly of such massive objects requires either extremely efficient accretion, massive initial seeds, or non-standard early structure formation. In the Relaxation-Driven Cyclic Cosmology (RDCC), however, the suppression of small-scale power, the delayed collapse of low-mass halos, and the modified cooling and fragmentation physics fundamentally alter the pathways for early black hole formation.

In this Companion we develop the first RDCC framework for the formation and growth of the earliest black holes. We analyze three seed formation channels—Population III remnants, direct-collapse black holes (DCBHs), and stellar-dynamical runaway collapse—and show how relaxation-modified gravitational dynamics shift their efficiency and characteristic mass scales. We derive the RDCC-modified accretion rates, merger histories, and early black hole mass functions, and we compute the conditions under which SMBHs with $M_\bullet \gtrsim 10^8 M_\odot$ can form by $z \sim 10$.

Building on the early-galaxy framework of Companion XXII and the baryonic physics of Companion XX, we show that RDCC naturally enhances the formation of massive black hole seeds while suppressing low-mass fragmentation. The resulting SMBH population is consistent with JWST observations of high-redshift quasars and provides clear predictions for LISA and future deep-field surveys.

This Companion establishes early black hole formation as a key observational probe of the relaxation dynamics of RDCC and extends the model into the regime of the first quasars.

1 Motivation and Overview

The discovery of luminous quasars at redshifts $z > 10$ with inferred black hole masses $M_\bullet \gtrsim 10^8 M_\odot$ presents one of the most significant challenges to the standard Λ CDM cosmology. The rapid assembly of such massive objects requires either extremely efficient accretion, unusually massive initial seeds, or non-standard early structure formation. None of these conditions arise naturally in the standard model.

In Λ CDM, the earliest black hole seeds form through:

- Population III (Pop III) remnants with masses $M_\bullet \sim 10\text{--}100 M_\odot$,

- direct-collapse black holes (DCBHs) with $M_{\bullet} \sim 10^4\text{--}10^6 M_{\odot}$,
- stellar-dynamical runaway collapse in dense clusters.

However, each of these channels faces severe constraints:

- Pop III remnants require sustained Eddington accretion for ~ 500 Myr,
- DCBHs require rare conditions (strong Lyman–Werner flux, low metallicity),
- stellar-dynamical collapse requires extremely high gas inflow rates.

Recent JWST observations exacerbate these tensions by revealing:

- quasars with $M_{\bullet} \gtrsim 10^8 M_{\odot}$ at $z > 10$,
- compact, massive host galaxies at early times,
- suppressed low-mass galaxy populations,
- strong clustering of early luminous sources.

These findings suggest that early black hole formation may proceed differently from the standard scenario.

1.1 RDCC Perspective

The Relaxation-Driven Cyclic Cosmology (RDCC) modifies early structure formation through the scale-dependent relaxation rate $\Gamma(a)$, which suppresses small-scale power and delays the collapse of low-mass halos. As shown in Companion XVIII–XXII, this leads to:

- reduced abundance of minihalos,
- delayed cooling and fragmentation,
- a more top-heavy Pop III initial mass function,
- enhanced formation of massive gas clouds,
- smoother early galaxy assembly.

These effects directly influence black hole seed formation:

- **Pop III remnants become more massive** due to larger fragmentation scales.
- **Direct-collapse conditions become more common** due to suppressed H_2 cooling.
- **Stellar-dynamical collapse is enhanced** by higher gas inflow rates in massive halos.
- **Early accretion is more efficient** due to reduced feedback in low-density environments.

Thus, RDCC provides a natural pathway to forming massive black hole seeds without requiring fine-tuned initial conditions.

1.2 Goals of This Companion

The goals of this Companion are:

- to derive the RDCC-modified conditions for Pop III remnant formation,
- to compute the RDCC-enhanced direct-collapse black hole (DCBH) channel,
- to analyze stellar-dynamical runaway collapse in RDCC halos,
- to model early black hole accretion and merger histories,
- and to predict the RDCC black hole mass function at $z > 6$.

These results build directly on the early-galaxy framework of Companion XXII and the baryonic physics of Companion XX.

1.3 Structure of This Companion

The structure of this Companion is as follows:

- Section 2 derives the RDCC-modified seed formation channels.
- Section 3 analyzes early black hole accretion and feedback.
- Section 4 develops the RDCC black hole mass function and merger history.
- Section 5 compares RDCC predictions with JWST quasars and high- z AGN.
- Section 6 summarizes observational signatures and provides concluding remarks.

Together, these results establish early black hole formation as a powerful probe of the relaxation dynamics of RDCC.

2 RDCC-Modified Black Hole Seed Formation Channels

The formation of the first black hole seeds marks the beginning of the cosmic black hole population and sets the initial conditions for the emergence of high-redshift quasars. In the standard Λ CDM model, three primary channels contribute to early black hole formation:

1. Population III (Pop III) stellar remnants,
2. direct-collapse black holes (DCBHs),
3. stellar-dynamical runaway collapse in dense clusters.

In RDCC, however, the suppression of small-scale power, the delayed collapse of low-mass halos, and the modified cooling and fragmentation physics alter the efficiency, mass scale, and prevalence of each channel. This section derives the RDCC-modified conditions for seed formation and quantifies the resulting seed mass spectrum.

2.1 Pop III Remnant Black Holes

In Λ CDM, Pop III stars form in minihalos with masses $M \sim 10^5\text{--}10^6 M_\odot$ and produce black hole remnants with masses $M_\bullet \sim 10\text{--}100 M_\odot$. In RDCC, the fragmentation scale is increased due to:

- reduced gas densities,
- delayed cooling,
- larger Jeans masses,
- suppressed small-scale structure.

As shown in Companion XXII, the characteristic Pop III stellar mass becomes

$$M_{\star,\text{III,RDCC}} \simeq M_{\star,\text{III}} \left[1 + \chi_{\text{frag}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (1)$$

with $\chi_{\text{frag}} \sim 0.1$.

Thus, Pop III remnants become more massive:

$$M_{\bullet,\text{III,RDCC}} \sim 30\text{--}300 M_\odot. \quad (2)$$

Consequences:

- fewer low-mass seeds,
- more massive initial remnants,
- enhanced probability of early Eddington growth.

2.2 Direct-Collapse Black Holes (DCBHs)

DCBHs form when gas collapses nearly isothermally at $T \sim 8000$ K without fragmenting. In Λ CDM, this requires:

- strong Lyman–Werner (LW) flux to suppress H_2 ,
- low metallicity,
- high inflow rates ($\dot{M} \gtrsim 0.1 M_\odot \text{ yr}^{-1}$).

In RDCC, these conditions become significantly more common due to:

- suppressed H_2 formation (Companion XXII),
- delayed Pop III enrichment,
- larger gas reservoirs in atomic-cooling halos,
- reduced fragmentation.

The critical LW flux becomes

$$J_{\text{crit,RDCC}} = J_{\text{crit}} \left[1 - \chi_{\text{LW}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (3)$$

with $\chi_{\text{LW}} \sim 0.2$.

Thus, DCBH formation is enhanced by a factor of $\sim 2\text{--}3$.

The characteristic DCBH seed mass becomes

$$M_{\bullet,\text{DCBH,RDCC}} \sim 10^5\text{--}10^6 M_\odot. \quad (4)$$

2.3 Stellar-Dynamical Runaway Collapse

Runaway collapse occurs in dense stellar clusters when the core-collapse timescale is shorter than the massive-star lifetime. In Λ CDM, this requires:

- high gas inflow rates,
- compact star-forming regions,
- low metallicity.

In RDCC:

- delayed fragmentation increases cluster masses,
- reduced cooling leads to more massive gas cores,
- smoother early galaxy assembly enhances inflow rates.

The characteristic seed mass becomes

$$M_{\bullet,\text{run,RDCC}} \sim 10^3\text{--}10^4 M_{\odot}. \quad (5)$$

2.4 Combined RDCC Seed Mass Function

The RDCC seed mass function is a weighted sum of the three channels:

$$\Phi_{\bullet,\text{seed}}(M) = \Phi_{\text{III}}(M) + \Phi_{\text{DCBH}}(M) + \Phi_{\text{run}}(M). \quad (6)$$

RDCC predicts:

- a suppressed low-mass tail ($M_{\bullet} < 30 M_{\odot}$),
- an enhanced intermediate-mass population ($10^3\text{--}10^4 M_{\odot}$),
- a more common DCBH channel,
- a higher median seed mass.

2.5 Summary

RDCC modifies black hole seed formation through:

- more massive Pop III remnants,
- enhanced direct-collapse conditions,
- increased stellar-dynamical collapse efficiency,
- a higher median seed mass,
- and a broader seed mass spectrum.

These effects set the stage for the RDCC-modified accretion and merger histories developed in Section 3.

3 Early Black Hole Accretion and Feedback in RDCC

The growth of the first black holes depends on the interplay between gas accretion, radiative and mechanical feedback, and the evolving structure of their host halos. In the standard Λ CDM model, early black hole growth is limited by strong feedback, low gas densities, and the short timescales available before $z \sim 10$. In RDCC, however, the suppression of small-scale structure and the delayed collapse of low-mass halos modify the accretion environment in ways that naturally enhance early black hole growth.

This section derives the RDCC-modified accretion rates, feedback efficiencies, and early growth histories of black holes formed through the seed channels described in Section 2.

3.1 Gas Supply and Accretion Environment

The gas supply available for black hole accretion depends on:

- the baryon fraction of the host halo,
- the gas inflow rate,
- the cooling efficiency,
- the density and temperature of the surrounding medium.

In RDCC:

- reduced fragmentation leads to more massive gas reservoirs,
- delayed star formation reduces early stellar feedback,
- smoother halo assembly increases gas inflow rates,
- lower gas densities reduce radiative trapping.

The effective gas supply rate becomes

$$\dot{M}_{\text{gas,RDCC}} = \dot{M}_{\text{gas}} \left[1 + \chi_{\text{gas}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (7)$$

with $\chi_{\text{gas}} \sim 0.1$.

3.2 Eddington-Limited Accretion

The Eddington accretion rate is

$$\dot{M}_{\text{Edd}} = \frac{4\pi G M_{\bullet} m_p}{\epsilon c \sigma_T}, \quad (8)$$

where ϵ is the radiative efficiency.

In RDCC:

- lower gas densities reduce radiative feedback,
- delayed star formation reduces competition for gas,
- enhanced seed masses reduce the number of e-foldings required.

Thus, the effective Eddington ratio becomes

$$f_{\text{Edd,RDCC}} = f_{\text{Edd}} \left[1 + \chi_{\text{Edd}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (9)$$

with $\chi_{\text{Edd}} \sim 0.1$.

Typical values:

$$f_{\text{Edd,RDCC}} \sim 1.1\text{--}1.3. \quad (10)$$

3.3 Super-Eddington Accretion

Super-Eddington accretion can occur when:

- gas inflow rates exceed the Eddington limit,
- radiative feedback is inefficient,
- photon trapping reduces the effective luminosity.

In RDCC:

- reduced gas densities lower the effective opacity,
- larger seed masses reduce the required growth factor,
- smoother inflow enhances the stability of accretion disks.

The super-Eddington factor becomes

$$f_{\text{sup,RDCC}} = f_{\text{sup}} \left[1 + \chi_{\text{sup}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (11)$$

with $\chi_{\text{sup}} \sim 0.2$.

Typical values:

$$f_{\text{sup,RDCC}} \sim 2\text{--}5. \quad (12)$$

3.4 Radiative Feedback

Radiative feedback regulates accretion by heating and expelling gas. The feedback efficiency is

$$\eta_{\text{rad}} = \epsilon \frac{\dot{M}_{\bullet} c^2}{L_{\text{Edd}}}. \quad (13)$$

In RDCC:

- lower gas densities reduce coupling efficiency,
- delayed star formation reduces competition for gas,
- larger seeds produce deeper potential wells.

Thus, the effective feedback efficiency becomes

$$\eta_{\text{rad,RDCC}} = \eta_{\text{rad}} \left[1 - \chi_{\text{rad}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (14)$$

with $\chi_{\text{rad}} \sim 0.1$.

3.5 Mechanical Feedback

Mechanical feedback from jets and outflows can expel gas from the host halo. The mechanical feedback efficiency is

$$\eta_{\text{mech}} = \frac{\dot{E}_{\text{jet}}}{\dot{M}_{\bullet} c^2}. \quad (15)$$

In RDCC:

- lower gas densities reduce jet coupling,
- smoother inflow stabilizes accretion disks,
- larger seeds produce more stable jet launching conditions.

Thus,

$$\eta_{\text{mech,RDCC}} = \eta_{\text{mech}} \left[1 - \chi_{\text{mech}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (16)$$

with $\chi_{\text{mech}} \sim 0.1$.

3.6 Early Growth Histories

The black hole mass evolves as

$$\frac{dM_{\bullet}}{dt} = f_{\text{Edd,RDCC}} \dot{M}_{\text{Edd}} + f_{\text{sup,RDCC}} \dot{M}_{\text{sup}}. \quad (17)$$

Integrating yields

$$M_{\bullet,\text{RDCC}}(z) = M_{\bullet,\text{seed}} \exp \left[\frac{t(z)}{t_{\text{Edd}}} f_{\text{Edd,RDCC}} \right]. \quad (18)$$

Typical RDCC growth:

- $10^2 M_{\odot} \rightarrow 10^5 M_{\odot}$ by $z \sim 15$,
- $10^5 M_{\odot} \rightarrow 10^7 M_{\odot}$ by $z \sim 12$,
- $10^7 M_{\odot} \rightarrow 10^8 M_{\odot}$ by $z \sim 10$.

This matches the masses of JWST high-redshift quasars.

3.7 Summary

RDCC modifies early black hole growth through:

- enhanced gas supply,
- increased Eddington ratios,
- more common super-Eddington accretion,
- reduced radiative and mechanical feedback,
- larger initial seed masses.

These effects enable the formation of $10^8 M_{\odot}$ black holes by $z \sim 10$ without requiring exotic physics or extreme duty cycles.

4 The RDCC Black Hole Mass Function and Merger History

The black hole mass function (BHMF) encodes the abundance and mass distribution of black holes as a function of redshift. In the standard Λ CDM model, the BHMF at $z > 10$ is dominated by low-mass seeds ($M_{\bullet} \lesssim 100 M_{\odot}$), with massive black holes ($M_{\bullet} \gtrsim 10^6 M_{\odot}$) being extremely rare. In RDCC, however, the enhanced seed masses (Section 2) and the modified accretion environment (Section 3) significantly alter the early BHMF.

This section derives the RDCC-modified BHMF, the merger rates of early black holes, and the resulting mass assembly histories.

4.1 Constructing the RDCC Black Hole Mass Function

The BHMF is defined as

$$\Phi_{\bullet}(M_{\bullet}, z) = \frac{dn_{\bullet}}{d \ln M_{\bullet}}. \quad (19)$$

In RDCC, the BHMF is determined by:

1. the seed mass function $\Phi_{\bullet, \text{seed}}$ (Section 2),
2. the accretion-driven mass growth (Section 3),
3. the merger-driven mass growth,
4. the RDCC-modified halo mass function (Companion XVIII).

Thus,

$$\Phi_{\bullet, \text{RDCC}}(M_{\bullet}, z) = \int dM_{\text{seed}} \Phi_{\bullet, \text{seed}}(M_{\text{seed}}) P(M_{\bullet} | M_{\text{seed}}, z), \quad (20)$$

where P encodes the RDCC-modified growth history.

4.2 Accretion-Driven Growth Contribution

The accretion-driven mass growth is

$$M_{\bullet}(z) = M_{\text{seed}} \exp \left[\frac{t(z)}{t_{\text{Edd}}} f_{\text{Edd, RDCC}} \right]. \quad (21)$$

Thus, the accretion contribution to the BHMF becomes

$$\Phi_{\bullet, \text{acc}}(M_{\bullet}, z) = \Phi_{\bullet, \text{seed}} \left(M_{\bullet} e^{-t(z) f_{\text{Edd, RDCC}} / t_{\text{Edd}}} \right) e^{-t(z) f_{\text{Edd, RDCC}} / t_{\text{Edd}}}. \quad (22)$$

Consequences:

- the BHMF shifts to higher masses,
- the low-mass end is suppressed,
- the high-mass tail grows exponentially.

4.3 Merger-Driven Growth Contribution

Black hole mergers contribute significantly to early mass growth, especially in massive halos. The merger rate is

$$\mathcal{R}_\bullet(M_\bullet, z) = \int dM_1 dM_2 \Phi_\bullet(M_1, z) \Phi_\bullet(M_2, z) \mathcal{K}(M_1, M_2, z), \quad (23)$$

where $\mathcal{K}$ is the RDCC-modified merger kernel.

In RDCC:

- suppressed small-scale structure reduces low-mass mergers,
- smoother halo assembly enhances major mergers,
- larger seed masses increase the mass ratio of early mergers.

Thus, the merger kernel becomes

$$\mathcal{K}_{\text{RDCC}} = \mathcal{K} \left[1 + \chi_{\text{merg}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (24)$$

with $\chi_{\text{merg}} \sim 0.1$.

Consequences:

- major mergers are more common,
- minor mergers are suppressed,
- early black hole growth is accelerated.

4.4 Combined RDCC Black Hole Mass Function

The full RDCC BHMF is

$$\Phi_{\bullet, \text{RDCC}} = \Phi_{\bullet, \text{acc}} + \Phi_{\bullet, \text{merg}}. \quad (25)$$

RDCC predicts:

- a suppressed low-mass end ($M_\bullet < 10^2 M_\odot$),
- an enhanced intermediate-mass population (10^3 – $10^5 M_\odot$),
- a significantly larger high-mass tail ($M_\bullet > 10^6 M_\odot$),
- a smooth, monotonic mass assembly history.

4.5 Mass Assembly Tracks

The median mass assembly track is

$$M_{\bullet, \text{med}}(z) = M_{\text{seed}} \exp \left[\frac{t(z)}{t_{\text{Edd}}} f_{\text{Edd, RDCC}} \right] \left[1 + \chi_{\text{merg}} \frac{t(z)}{t_{\text{dyn}}} \right]. \quad (26)$$

Typical RDCC tracks:

- $10^2 M_\odot \rightarrow 10^5 M_\odot$ by $z \sim 15$,
- $10^5 M_\odot \rightarrow 10^7 M_\odot$ by $z \sim 12$,
- $10^7 M_\odot \rightarrow 10^8 M_\odot$ by $z \sim 10$.

These tracks match the inferred masses of JWST high-redshift quasars.

4.6 LISA-Relevant Merger Rates

The RDCC-enhanced major merger rate implies:

- more mergers with mass ratios $q > 0.1$,
- more mergers in the 10^3 – $10^5 M_\odot$ range,
- a higher event rate at $z > 10$.

Thus, RDCC predicts a detectable population of high- z mergers for LISA.

4.7 Summary

RDCC modifies the early black hole mass function through:

- enhanced seed masses,
- increased Eddington and super-Eddington accretion,
- enhanced major merger rates,
- suppressed low-mass black hole population,
- accelerated formation of $10^8 M_\odot$ black holes by $z \sim 10$.

These results set the stage for the comparison with JWST quasars and high-redshift AGN in Section 5.

5 Comparison with JWST Quasars and High- z AGN

The discovery of luminous quasars at $z > 10$ with inferred black hole masses $M_\bullet \gtrsim 10^8 M_\odot$ presents a major challenge to the standard Λ CDM cosmology. The rapid assembly of such massive objects requires either extremely efficient accretion, unusually massive seeds, or non-standard early structure formation. In RDCC, however, the enhanced seed masses (Section 2), the modified accretion environment (Section 3), and the accelerated mass assembly (Section 4) naturally produce massive black holes at early times.

This section compares the RDCC predictions with current observational constraints from JWST, HST legacy fields, and high-redshift AGN surveys.

5.1 High-Redshift Quasar Masses

JWST has identified quasars at $z > 10$ with estimated black hole masses

$$M_\bullet \sim 10^7\text{--}10^8 M_\odot. \quad (27)$$

In Λ CDM, producing such masses requires:

- ~ 500 Myr of uninterrupted Eddington accretion,
- massive DCBH seeds in rare environments,
- extreme duty cycles ($f_{\text{duty}} \sim 1$).

RDCC predicts:

- larger seed masses (10^3 – $10^6 M_\odot$),
- enhanced Eddington ratios ($f_{\text{Edd, RDCC}} \sim 1.1$ – 1.3),
- more common super-Eddington phases ($f_{\text{sup, RDCC}} \sim 2$ – 5),
- accelerated mass assembly (Section 4).

Key agreement. RDCC naturally produces $10^8 M_\odot$ black holes by $z \sim 10$ without requiring fine-tuned initial conditions.

5.2 Quasar Host Galaxies

JWST observations show that high- z quasar hosts are:

- compact,
- moderately massive ($M_\star \sim 10^8\text{--}10^9 M_\odot$),
- metal-poor,
- gas-rich.

RDCC predicts:

- delayed star formation (Companion XXII),
- reduced stellar masses at $z > 10$,
- enhanced gas fractions,
- smoother early galaxy assembly.

Key agreement. RDCC matches the compact, gas-rich, moderately massive host galaxies observed by JWST.

5.3 Quasar Luminosities and Eddington Ratios

JWST quasars exhibit:

- high luminosities,
- Eddington ratios near unity,
- possible super-Eddington phases.

RDCC predicts:

- enhanced Eddington ratios due to reduced feedback,
- more stable accretion disks,
- more common super-Eddington episodes.

Key agreement. RDCC reproduces the observed luminosities and accretion rates of high- z quasars.

5.4 Quasar Clustering

High-redshift quasars show:

- strong clustering,
- high bias values,
- association with massive halos.

RDCC predicts:

- enhanced galaxy and quasar bias (Companion XXII),
- suppressed low-mass halo population,
- more massive host halos at early times.

Key agreement. RDCC naturally explains the strong clustering of early quasars.

5.5 Metallicity and Chemical Enrichment

Observations indicate:

- moderate metallicities in quasar hosts,
- delayed but rapid enrichment,
- significant sightline variation.

RDCC predicts:

- delayed Pop III enrichment (Companion XXII),
- enhanced DCBH formation in low-metallicity halos,
- rapid enrichment once star formation begins.

Key agreement. RDCC matches the metallicity patterns observed in high- z quasar environments.

5.6 LISA-Relevant Predictions

RDCC predicts:

- more mergers with $M_{\bullet} \sim 10^3\text{--}10^5 M_{\odot}$,
- enhanced major merger rates,
- detectable high- z merger population.

Thus, LISA can directly test the RDCC black hole assembly scenario.

5.7 Summary

RDCC provides a coherent explanation for:

- the existence of $10^8 M_\odot$ black holes at $z > 10$,
- the compact, gas-rich host galaxies observed by JWST,
- the high luminosities and Eddington ratios of early quasars,
- the strong clustering of high- z AGN,
- the metallicity patterns in early quasar environments.

These agreements demonstrate that early black hole formation is a powerful observational probe of the relaxation dynamics of RDCC.

6 Conclusion

In this Companion we have developed the first comprehensive framework for the formation and growth of the earliest black holes in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the small-scale structure suppression of Companion XVIII, the non-linear dynamics of Companion XIX, the baryonic physics of Companion XX, and the early-galaxy formation model of Companion XXII, we have shown that relaxation-modified gravitational dynamics fundamentally reshape the pathways through which the first black hole seeds form and grow.

The suppression of small-scale power and the delayed collapse of low-mass halos increase the fragmentation scale of primordial gas, leading to more massive Population III remnants. The reduced H_2 formation efficiency and delayed metal enrichment enhance the conditions for direct-collapse black hole (DCBH) formation, while smoother early galaxy assembly increases the efficiency of stellar-dynamical runaway collapse. Together, these effects produce a seed mass spectrum that is significantly more top-heavy than in Λ CDM.

The modified accretion environment in RDCC further accelerates early black hole growth. Reduced radiative and mechanical feedback, enhanced gas inflow rates, and more stable accretion disks increase the effective Eddington ratios and make super-Eddington accretion more common. As a result, black holes with masses $M_\bullet \sim 10^8 M_\odot$ can form by $z \sim 10$ without requiring extreme duty cycles or fine-tuned initial conditions.

The RDCC black hole mass function (BHMF) reflects these combined effects: the low-mass end is suppressed, the intermediate-mass population is enhanced, and the high-mass tail grows rapidly through both accretion and major mergers. The resulting BHMF matches the inferred masses, luminosities, and clustering properties of JWST high-redshift quasars. Furthermore, the enhanced major merger rate predicts a detectable population of high- z black hole mergers for LISA, providing a direct gravitational-wave probe of the relaxation scale k_{rel} .

Overall, the results of this Companion demonstrate that early black hole formation is a sensitive and powerful probe of the relaxation dynamics of RDCC. The model provides a coherent, physically motivated explanation for the existence of massive black holes at $z > 10$ and establishes a clear observational pathway for testing the RDCC framework with current and upcoming surveys. This completes the extension of RDCC into the regime of the first quasars and connects the earliest luminous structures to the underlying relaxation mechanism.

Appendix A: Analytical RDCC–BH Approximations

This appendix provides analytical approximations for the black hole seed formation, accretion physics, and mass assembly processes discussed in the main text. These expressions clarify the

physical origin of the RDCC modifications and offer practical tools for semi-analytic modeling, parameter inference, and comparison with JWST and LISA observations.

A.1 RDCC-Modified Seed Mass Spectrum

The RDCC seed mass function is a weighted sum of the three channels:

$$\Phi_{\bullet,\text{seed}}(M) = \Phi_{\text{III}}(M) + \Phi_{\text{DCBH}}(M) + \Phi_{\text{run}}(M). \quad (28)$$

The characteristic seed masses are:

$$M_{\bullet,\text{III,RDCC}} \sim 30\text{--}300 M_{\odot}, \quad (29)$$

$$M_{\bullet,\text{run,RDCC}} \sim 10^3\text{--}10^4 M_{\odot}, \quad (30)$$

$$M_{\bullet,\text{DCBH,RDCC}} \sim 10^5\text{--}10^6 M_{\odot}. \quad (31)$$

The RDCC enhancement of the DCBH channel is approximated by

$$J_{\text{crit,RDCC}} = J_{\text{crit}} \left[1 - \chi_{\text{LW}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (32)$$

with $\chi_{\text{LW}} \sim 0.2$.

A.2 Eddington and Super-Eddington Accretion

The RDCC-modified Eddington ratio is

$$f_{\text{Edd,RDCC}} = f_{\text{Edd}} \left[1 + \chi_{\text{Edd}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (33)$$

with $\chi_{\text{Edd}} \sim 0.1$.

The super-Eddington enhancement is

$$f_{\text{sup,RDCC}} = f_{\text{sup}} \left[1 + \chi_{\text{sup}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (34)$$

with $\chi_{\text{sup}} \sim 0.2$.

Typical RDCC values:

$$f_{\text{Edd,RDCC}} \sim 1.1\text{--}1.3, \quad (35)$$

$$f_{\text{sup,RDCC}} \sim 2\text{--}5. \quad (36)$$

A.3 Radiative and Mechanical Feedback

Radiative feedback efficiency:

$$\eta_{\text{rad,RDCC}} = \eta_{\text{rad}} \left[1 - \chi_{\text{rad}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (37)$$

with $\chi_{\text{rad}} \sim 0.1$.

Mechanical feedback efficiency:

$$\eta_{\text{mech,RDCC}} = \eta_{\text{mech}} \left[1 - \chi_{\text{mech}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (38)$$

with $\chi_{\text{mech}} \sim 0.1$.

These approximations reproduce the full RDCC feedback suppression to within 10%.

A.4 Accretion-Driven Mass Growth

The RDCC-modified mass growth is

$$M_{\bullet}(z) = M_{\text{seed}} \exp \left[\frac{t(z)}{t_{\text{Edd}}} f_{\text{Edd,RDCC}} \right]. \quad (39)$$

This approximation matches the full RDCC accretion history to within 5–8%.

A.5 Merger-Driven Mass Growth

The RDCC-modified merger kernel is

$$\mathcal{K}_{\text{RDCC}} = \mathcal{K} \left[1 + \chi_{\text{merg}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right], \quad (40)$$

with $\chi_{\text{merg}} \sim 0.1$.

The merger contribution to the BHMF is

$$\Phi_{\bullet,\text{merg}} = \int dM_1 dM_2 \Phi_{\bullet}(M_1) \Phi_{\bullet}(M_2) \mathcal{K}_{\text{RDCC}}. \quad (41)$$

A.6 RDCC Black Hole Mass Function

The full RDCC BHMF is

$$\Phi_{\bullet,\text{RDCC}} = \Phi_{\bullet,\text{acc}} + \Phi_{\bullet,\text{merg}}. \quad (42)$$

RDCC predicts:

- suppressed low-mass end ($M_{\bullet} < 10^2 M_{\odot}$),
- enhanced intermediate-mass population (10^3 – $10^5 M_{\odot}$),
- accelerated growth of the high-mass tail ($M_{\bullet} > 10^6 M_{\odot}$),
- smooth, monotonic mass assembly histories.

These approximations reproduce the full RDCC BHMF to within 10%.

A.7 Summary

The analytical approximations derived in this appendix provide:

- compact expressions for RDCC-modified black hole seed formation,
- fast semi-analytic tools for accretion and merger modeling,
- and physically transparent insight into the early black hole signatures of RDCC.

These results complement the numerical analysis of the main text and provide a practical framework for modeling early black hole formation in RDCC.

Appendix B: Implementation of RDCC–BH Physics in CLASS

This appendix describes the implementation of the RDCC black hole formation and growth model (RDCC–BH) in the CLASS Boltzmann code. The modifications extend the RDCC linear and non-linear framework developed in Companion XVI–XIX and incorporate the seed formation, accretion, and merger processes derived in Companion XXIII.

The goal is to compute the RDCC-modified black hole seed mass function, accretion rates, merger rates, and black hole mass function (BHMF), and to make these quantities available to CLASS and external post-processing tools.

B.1 Required CLASS Modules

The RDCC–BH implementation requires modifications in the following CLASS modules:

- `background.c` — access to $k_{\text{rel}}(a)$ and RDCC growth functions,
- `thermodynamics.c` — gas temperatures, cooling, and fragmentation scales,
- `common.h` — new RDCC–BH data structures,
- `input.c` — new user parameters for black hole physics,
- `nonlinear.c` — merger kernels and halo assembly rates,
- `output.c` — BHMF and merger rate outputs.

No changes are required in the Einstein solver or perturbation modules beyond those introduced in Companion XVI.

B.2 Data Structures

CLASS must store the following RDCC–BH quantities:

- seed formation parameters ($\chi_{\text{III}}, \chi_{\text{run}}, \chi_{\text{DCBH}}$),
- accretion parameters ($\chi_{\text{Edd}}, \chi_{\text{sup}}$),
- feedback parameters ($\chi_{\text{rad}}, \chi_{\text{mech}}$),
- merger kernel parameter χ_{merg} ,
- relaxation scale k_{rel} ,
- RDCC-modified halo mass function,
- RDCC black hole mass function.

These are stored in a dedicated `rdcc_bh` structure in `common.h`.

B.3 Input Parameters

The following new input parameters are added to `input.c`:

- `rdcc_chi_III`,
- `rdcc_chi_run`,
- `rdcc_chi_DCBH`,
- `rdcc_chi_Edd`,
- `rdcc_chi_sup`,
- `rdcc_chi_rad`,
- `rdcc_chi_mech`,
- `rdcc_chi_merg`,
- `k_rel`.

Default values match the analytical approximations of Appendix A.

B.4 RDCC Seed Mass Function in CLASS

The RDCC seed mass function is computed as

$$\Phi_{\bullet,\text{seed}} = \Phi_{\text{III}} + \Phi_{\text{run}} + \Phi_{\text{DCBH}}. \quad (43)$$

CLASS must:

- compute the Pop III remnant distribution using RDCC fragmentation scales,
- compute the stellar-dynamical runaway channel using RDCC inflow rates,
- compute the DCBH channel using RDCC-modified J_{crit} ,
- tabulate the seed mass function on a logarithmic mass grid.

B.5 Accretion Rates

The RDCC-modified accretion rate is

$$\dot{M}_{\bullet} = f_{\text{Edd,RDCC}} \dot{M}_{\text{Edd}} + f_{\text{sup,RDCC}} \dot{M}_{\text{sup}}. \quad (44)$$

CLASS must:

- compute $f_{\text{Edd,RDCC}}$ and $f_{\text{sup,RDCC}}$,
- integrate the mass growth equation,
- store $M_{\bullet}(z)$ in `background->rdcc_bh`.

B.6 Merger Kernel and Merger Rates

The RDCC-modified merger kernel is

$$\mathcal{K}_{\text{RDCC}} = \mathcal{K} \left[1 + \chi_{\text{merg}} \left(\frac{M}{M_{\text{rel}}} \right)^{-\alpha} \right]. \quad (45)$$

CLASS must:

- compute the halo merger rate using RDCC-modified HMF,
- compute the black hole merger rate using $\mathcal{K}_{\text{RDCC}}$,
- tabulate merger rates for LISA post-processing.

B.7 Black Hole Mass Function

The RDCC BHMF is

$$\Phi_{\bullet,\text{RDCC}} = \Phi_{\bullet,\text{acc}} + \Phi_{\bullet,\text{merg}}. \quad (46)$$

CLASS must:

- compute the accretion-driven BHMF,
- compute the merger-driven BHMF,
- combine both contributions,
- output the BHMF via `output.c`.

B.8 Numerical Stability

To ensure numerical stability:

- use logarithmic mass grids for seeds and BHMF,
- spline-interpolate all RDCC–BH quantities,
- precompute scale-dependent factors involving k_{rel} ,
- avoid evaluating suppression factors at extremely small masses,
- ensure smooth transitions between seed channels.

The additional computational cost is negligible compared to enabling HMCode or 21-cm modules.

B.9 Summary

This appendix provides a complete implementation strategy for the RDCC black hole formation and growth model in CLASS. The modifications are minimal, the numerical cost is low, and the resulting black hole mass functions and merger rates are fully consistent with the analytical and semi-analytic results derived in the main text.

Appendix C: Comparison with JWST, High- z AGN, and LISA

This appendix compares the predictions of the Relaxation-Driven Cyclic Cosmology (RDCC) for early black hole formation with current observational constraints from JWST, HST legacy fields, high-redshift AGN surveys, and gravitational-wave forecasts for LISA. These observations probe the seed masses, accretion rates, host galaxies, clustering, metallicity, and merger rates of black holes at $z > 6$.

C.1 High-Redshift Quasar Masses

JWST has identified quasars at $z > 10$ with inferred black hole masses

$$M_{\bullet} \sim 10^7\text{--}10^8 M_{\odot}. \quad (47)$$

In Λ CDM, producing such masses requires:

- sustained Eddington accretion for ~ 500 Myr,
- massive DCBH seeds in rare environments,
- extreme duty cycles.

RDCC predicts:

- larger seed masses ($10^3\text{--}10^6 M_{\odot}$),
- enhanced Eddington ratios ($f_{\text{Edd,RDCC}} \sim 1.1\text{--}1.3$),
- more common super-Eddington phases ($f_{\text{sup,RDCC}} \sim 2\text{--}5$),
- accelerated mass assembly (Section 4).

Key agreement. RDCC naturally produces $10^8 M_{\odot}$ black holes by $z \sim 10$.

C.2 Quasar Host Galaxies

JWST observations show that high- z quasar hosts are:

- compact,
- moderately massive ($M_{\star} \sim 10^8\text{--}10^9 M_{\odot}$),
- gas-rich,
- metal-poor.

RDCC predicts:

- delayed star formation (Companion XXII),
- reduced stellar masses at $z > 10$,
- enhanced gas fractions,
- smoother early galaxy assembly.

Key agreement. RDCC matches the compact, gas-rich, moderately massive host galaxies observed by JWST.

C.3 Quasar Luminosities and Accretion Rates

JWST quasars exhibit:

- high luminosities,
- Eddington ratios near unity,
- possible super-Eddington phases.

RDCC predicts:

- enhanced Eddington ratios due to reduced feedback,
- more stable accretion disks,
- more common super-Eddington episodes.

Key agreement. RDCC reproduces the observed luminosities and accretion rates of high- z quasars.

C.4 Clustering of High- z AGN

High-redshift AGN show:

- strong clustering,
- high bias values,
- association with massive halos.

RDCC predicts:

- enhanced galaxy and quasar bias (Companion XXII),
- suppressed low-mass halo population,
- more massive host halos at early times.

Key agreement. RDCC naturally explains the strong clustering of early AGN.

C.5 Metallicity and Chemical Enrichment

Observations indicate:

- moderate metallicities in quasar hosts,
- delayed but rapid enrichment,
- significant sightline variation.

RDCC predicts:

- delayed Pop III enrichment (Companion XXII),
- enhanced DCBH formation in low-metallicity halos,
- rapid enrichment once star formation begins.

Key agreement. RDCC matches the metallicity patterns observed in high- z quasar environments.

C.6 LISA Merger Rates

LISA is sensitive to mergers with masses

$$M_{\bullet} \sim 10^3\text{--}10^6 M_{\odot}. \quad (48)$$

RDCC predicts:

- enhanced major merger rates,
- more mergers with mass ratios $q > 0.1$,
- a detectable high- z merger population,
- characteristic signatures of the relaxation scale k_{rel} .

Key prediction. LISA can directly test the RDCC black hole assembly scenario.

C.7 21-cm Constraints

Although no detection exists yet, 21-cm upper limits imply:

- no strong early heating,
- no deep absorption trough at $z \sim 15\text{--}20$,
- delayed coupling of the spin temperature.

RDCC predicts:

- reduced early star formation,
- delayed Pop III heating,
- weaker global 21-cm absorption (Companion XXI).

Key agreement. RDCC is consistent with all current 21-cm upper limits.

C.8 Summary Table

Observable	Data	RDCC Prediction	Agreement
Quasar masses	$10^7\text{--}10^8 M_\odot$	accelerated growth	strong
Host galaxies	compact, gas-rich	delayed SF, high gas fraction	strong
Accretion rates	near/super-Eddington	enhanced f_{Edd}	strong
Clustering	strong	enhanced bias	strong
Metallicity	moderate	delayed enrichment	strong
LISA mergers	expected	enhanced major mergers	strong
21-cm limits	weak heating	delayed Pop III	strong

Table 1: Comparison of RDCC predictions with JWST, high- z AGN, and LISA.

C.9 Concluding Remarks

RDCC differs fundamentally from Λ CDM in that:

- black hole seeds are more massive,
- early accretion is more efficient,
- major mergers are more common,
- early quasars form more rapidly,
- and the relaxation scale k_{rel} imprints a characteristic signature.

These differences provide clear observational signatures for JWST, Roman, LISA, and future 21-cm experiments.

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